

Activity Guide: Console Orientation — Projects, Datasets, and Tables

Guided follow-along on Day 1 (10:10–10:25 AM CT), right after M1: your first hands-on tour of the BigQuery console. Based on §3 ("Exploring projects, datasets, and tables") of the [Do-It-Now \(DIN\) activities page](#). No cost; everything happens in the console.

What you'll do

- Open the BigQuery console and pin it for the day
- Star three projects into your Explorer: bigquery-samples, bigquery-public-data, roi-bq-demos
- Explore a public dataset and a 100-billion-row table (schema, details, preview — no query, no cost)
- Create your own dataset named class — **required** for this afternoon's loading activities

Why it matters for your migration

- **Project → dataset → table** is the GCP resource hierarchy — the same shape as your **instance → database → schema.table**. Everything you build on GCP hangs off this skeleton.
- **Starring is your first taste of 'query data where it lives.'** Starred projects belong to someone else and stay read-only, yet you can query across them without copying a byte. That is the mechanic behind BigQuery as a single source of truth.
- **The class dataset is your landing zone.** Every load this afternoon lands there — think of it as your Bronze layer in miniature.
- **The 100-billion-row table is the scale story, felt.** You're about to browse 100B rows with zero servers provisioned.

Before you start

- You are signed in to the Google Cloud console with the lab user account from your [Skills classroom](#) lab.
- You know how to check which project is selected: the **project picker** in the top bar.

Steps

Follow your instructor's screen; the annotations below tell you what to notice at each stop.

1. Get to BigQuery

1. Open the Google Cloud console.
2. Open the navigation menu (the "hamburger" icon, top left) and choose **BigQuery** — or type "BigQuery" into the search bar at the top.

3. Hover over **BigQuery** in the navigation menu and click the pin, so it stays at the top all day.
4. In the **Explorer** pane (left), expand your project. Notice it's empty — nothing exists yet.

2. Star three projects

Starring pins someone else's project into your Explorer so you can browse and query it.

5. In the Explorer pane, click ♦ **Add**, then choose **Star a project by name**.
6. Type the name exactly and confirm: bigquery-samples
7. Repeat for: bigquery-public-data
8. Repeat for: roi-bq-demos
9. Star any additional projects your instructor calls out.

(Console docs: [starring a project](#))

3. Explore datasets

10. Expand **bigquery-samples** in the Explorer. Each entry under it is a **dataset** — a collection of tables and views that is secured as a unit and lives in a specific location.
11. Click **wikipedia_benchmark** and read the metadata shown in the right pane.
12. Click the three-dot menu (⋮) next to a dataset name and note the available actions. **This menu is where Create table lives** — you'll use it at 10:40.

4. Create your class dataset — required

13. Click the ⋮ menu next to **your own project** → **Create dataset**.
14. Dataset ID: class — exactly that, lowercase, no spaces. The other defaults are fine for class; note the **location** shown before you confirm.
15. Click **Create dataset**, then expand your project to confirm it appears.

Do not skip this step. The afternoon loading activities (DIN §24–§27) write into your class dataset. If it doesn't exist by 13:45, you'll be creating it while everyone else is loading data.

5. Explore tables

16. Expand **bigquery-samples** → **wikipedia_benchmark** and review its tables.
17. Click the **100B** table — yes, that's 100 billion rows.
18. **Schema** tab: column names and types — the same idea as the SSMS columns view or sp_help.
19. **Details** tab: storage size, row count, last-modified time.
20. **Preview** tab: sample rows *without running a query* — free browsing of a 100-billion-row table.
21. Open the ⋮ menu next to the table name and note the table-level actions.

Common gotchas

Wrong project selected. The console remembers your last-used project and happily shows it to you. If you "can't see what the instructor sees," check the **project picker in the top bar first**. This is the number-one cause of confusion today.

Starred projects are read-only. You cannot create anything inside bigquery-samples, bigquery-public-data, or roi-bq-demos — permission errors there are expected. Create things in **your** project.

Dataset location is permanent. You choose it at creation and cannot change it afterward — fixing it means recreating or copying the dataset. For class, the default is fine; for your real warehouse, it's an architecture decision.

The three-dot (⋮) menus hide the actions. Create dataset, Create table, Delete — if you can't find a button, look for the ⋮.

New dataset not showing? Expand your project node in the Explorer pane, or click the refresh icon at the top of the pane.

If you finish early

- Browse into bigquery-public-data, pick any dataset that looks interesting, and open a table's **Preview** tab.
- On the 100B table's **Details** tab, find the exact row count and total size.
- Open your own class dataset's ⋮ menu and locate **Create table** — your next stop at 10:40.

Self-check

- ☐ Three starred projects appear in your Explorer: bigquery-samples, bigquery-public-data, roi-bq-demos
- ☐ Your own project contains a dataset named exactly class
- ☐ You can open the Schema, Details, and Preview tabs on bigquery-samples.wikipedia_benchmark.100B
- ☐ You know where the project picker is, and which project is yours
- ☐ You found **Create table** in your dataset's ⋮ menu

What this replaces in your current stack

You do this today	On GCP
SSMS Object Explorer: server → databases → tables	BigQuery Explorer pane: projects → datasets → tables
Linked servers / browsing another instance	Starred projects — query across projects without copying data
CREATE DATABASE plus file and filegroup sizing	Create dataset — one dialog, nothing to size

Next: [Lab 1 companion — Loading Data into BigQuery](#)

Part of the Iowa DOM DoIT Data Engineering student pack — see ../00_schedule.md.